



LabClear 4.0.0-rc3

Book a health check. Understand your results.

A Thai-first chatbot for a simulated health-check clinic. It answers only from the clinic's own data, reads lab reports the customer confirms, and explains each value against the range printed on the same report, with sources.

นายวัชรินทร์ บัวสอน 68076055 · นายศิริพล ศรีเฮงไพบูลย์ 68076060

06048308 Intelligent Chatbot Development · Final Project · October 2026

Customers ask the same questions at every visit

When	What customers ask
Before	I have 1,500 baht. Which package should I take?
Booking	Which center is free? Can I move my appointment?
Results	What does HbA1c on my report mean? Which values are out of range?

LabClear (simulated business)

18 packages

290–2,890 baht · follow-up tests need staff review first

3 centers

Bangkok, Chiang Mai, Khon Kaen · Mon–Sat 07:00–16:00

AI Lab Report

one free report · Plus 355 baht per 30 days

Prices, centers, payments and lab reports are synthetic. No real patient data, no real money.

From the report a customer has to the check-up they need

- 1 Attach a photo or PDF of the report
Read in an isolated worker, then by Typhoon OCR, page by page
- 2 The customer checks and confirms the values
Nothing is explained before confirmation
- 3 Explanation with sources
Compared only with the range printed on the same report; no diagnosis
- 4 Ask about follow-up tests if wanted
The customer chooses; out-of-range values never trigger a sale
- 5 Request a time; staff confirm; then pay
Bookings and payments are previews until the customer confirms

รายการตรวจ ↓	ผลตรวจ	หน่วย	ช่วงบนรายงาน
Glucose	100	mg/dL	70-99

Values read from a sample report, waiting for confirmation (offline test double)

One FastAPI service on Render

Website and API together

Jinja pages and plain JavaScript, Thai first with an English switch

Chat pipeline

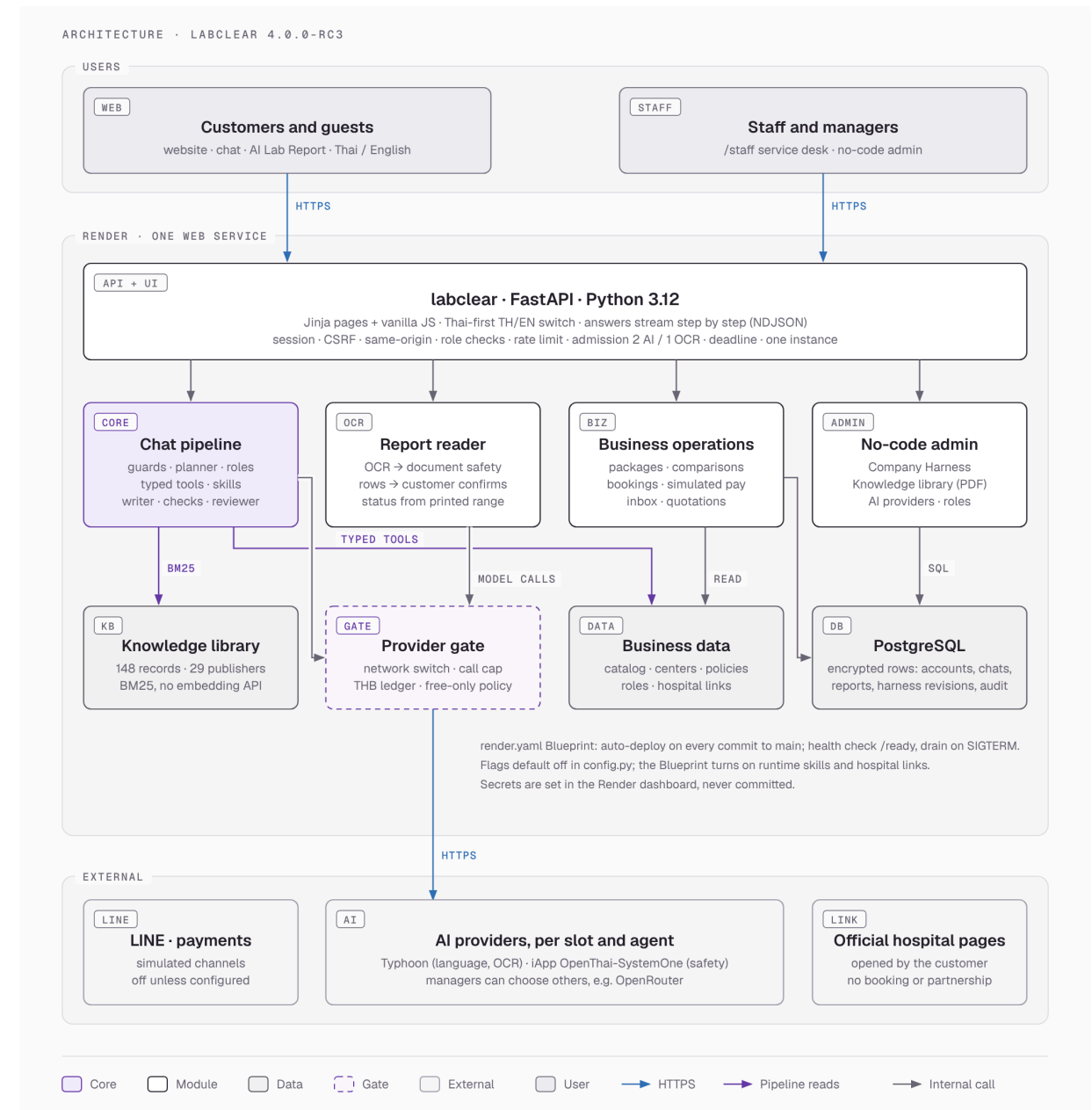
guards, planner, roles, typed tools, runtime skills, writer, checks, reviewer

Provider gate

network switch, call cap, THB ledger and a free-only policy before every model call

Blueprint

auto-deploys every commit on `main`; health check `/ready`



One message, step by step

The model proposes, Python decides
the planner suggests a role and search terms; code
checks permissions and calls the tools

Answers cite evidence

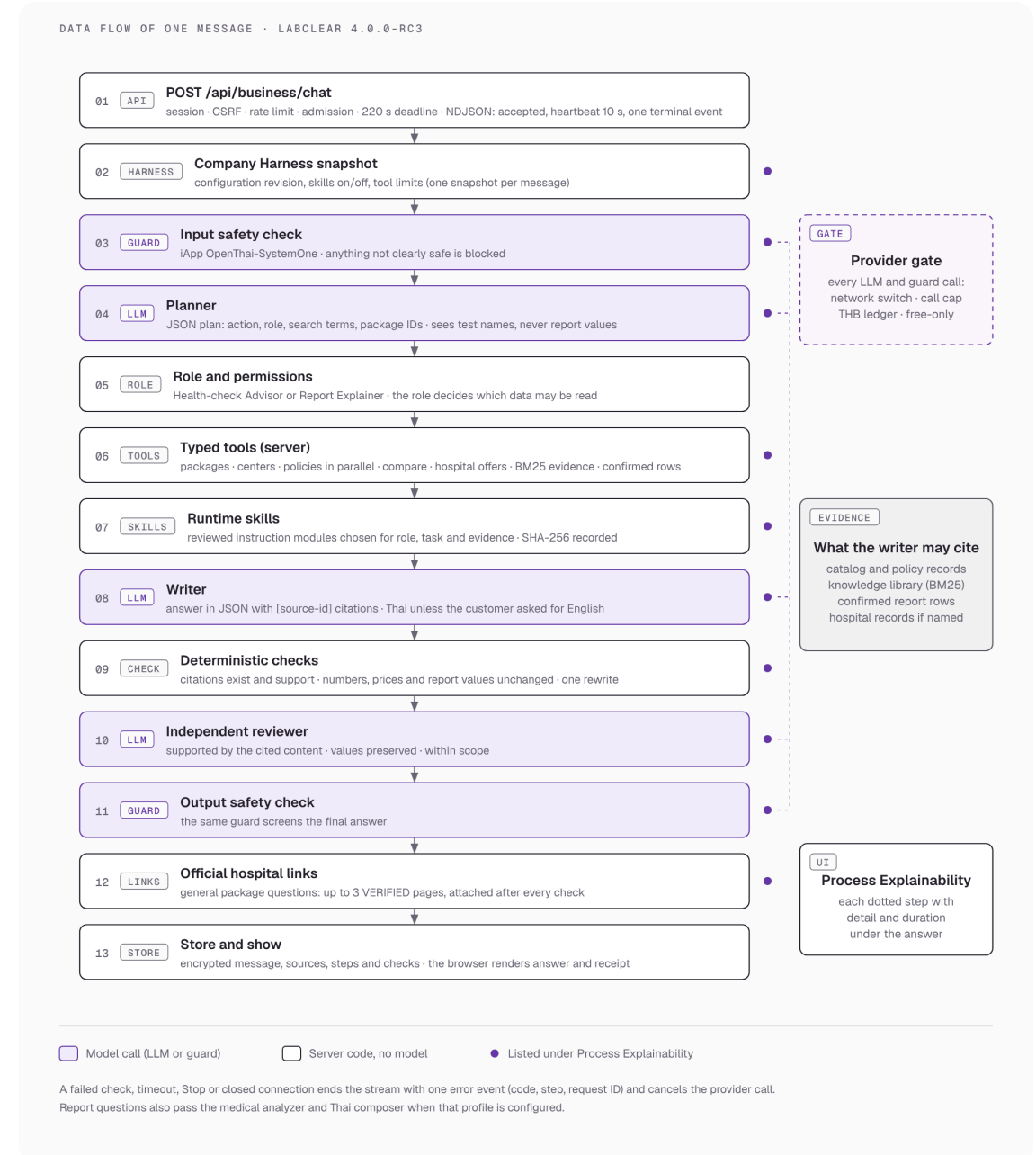
only sources the tools returned; prices and report
values checked in code; one rewrite, then withheld

Five model calls

input guard, planner, writer, reviewer, output guard

Every step is visible

streamed live and kept under the answer as Process
Explainability



Typed tools, reviewed skills, visible steps

Tool	Advisor	Explainer
lookup_packages · compare_packages	✓	—
lookup_branches	✓	—
lookup_policies	✓	✓
retrieve_evidence (BM25, 148 records)	✓	✓
get_confirmed_report_rows	—	✓
preview_booking · get_external_hospital_offer	✓	—

ดูแหล่งอ้างอิง 23 รายการ

Process Explainability

ถามทีมของเรา

- ✓ Company Harness
Typed tools labclear-tools-1.1.0 · การตั้งค่ารุ่น r0 · เปิด runtime skills
- ✓ คำถามผ่านการตรวจสอบความปลอดภัย
iApp OpenThai-SystemOne · 30.2 ms
- ✓ แผน: ตอบคำถาม โดยผู้ช่วยเลือกแพ็คเกจ
OFFLINE planner stand-in (not a language model) · 28.6 ms
- ✓ เรียกใช้ tool: lookup_packages
18 รายการ · 2.0 ms · 2 ms
- ✓ เรียกใช้ tool: lookup_branches
1 รายการ · 0.2 ms · 0.2 ms
- ✓ เรียกใช้ tool: lookup_policies
1 รายการ · 0.1 ms · 0.1 ms
- ✓ โหลดข้อมูลบริการที่ผู้ช่วยเลือกแพ็คเกจใช้ได้
catalog, centers, policies
- ✓ เรียกใช้ tool: retrieve_evidence

8 skill modules chosen per role and task, SHA-256 checked; every tool call audited without storing its arguments

Managers tune the assistant without code, inside safety rails

Company Harness

turn unlocked skills on or off, add company wording, sources per search 1–8, answer length 500–4,000 tokens

Limits can only go down

tool timeouts and item counts never exceed the code's ceiling

Locked

core, evidence-citation and scope-uncertainty; safety checks; data permissions; customer confirmation

Knowledge library

148 owner-approved records as A4 PDF pages; pause any record

Company Harness

เลือก skills ที่ผ่านการตรวจทานและ typed tools ที่ผู้ช่วยใช้ ทุกคำตอบจะบันทึก skills และ tools จริงไว้ใน "Process Explainability" การเปลี่ยนแปลงมีผลตั้งแต่ข้อความถัดไป

จำนวนแหล่งอ้างอิงต่อการค้นหา

ตั้งแต่ 1 ถึง 8

☒ ใช้ runtime skills

6

เปิดเสมอและแก้ไขในหน้านี้ไม่ได้: การตรวจความปลอดภัยของทุกข้อความและคำตอบ การตรวจสอบสิทธิ์เจ้าของ
จองทุกครั้ง

Runtime skills

ชุดคำสั่งที่ผ่านการตรวจทาน ระบบจะเลือกเฉพาะชุดที่เกี่ยวข้องกับแต่ละข้อความ คุณปิดชุดที่ไม่ต้องการ หรือ

Every save is a new revision that can be restored; its SHA-256 is recorded with each answer

A request can no longer hang or end in a raw 502

220 s one deadline for the whole chat request	10 s heartbeat while waiting for the model	2 / 1 AI / OCR requests at once; then 503 with Retry-After	12 / 12 fault cases R01–R12 passed, 183 assertions
---	--	--	--

Exactly one terminal event

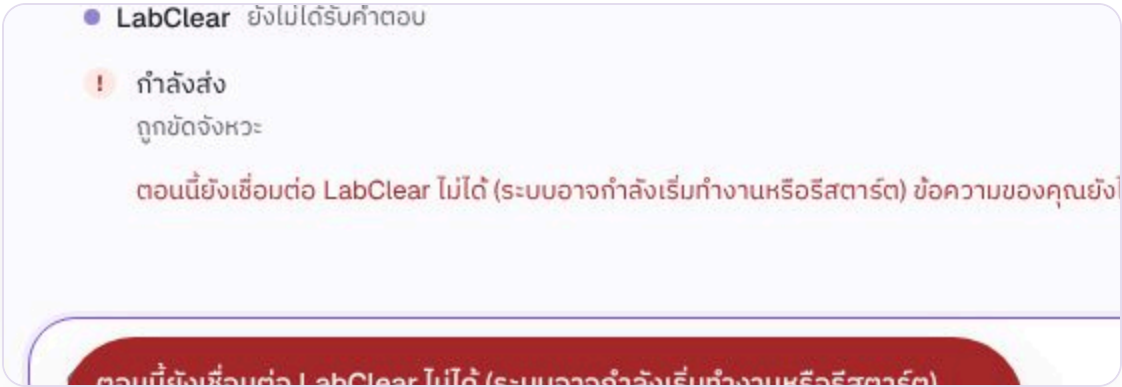
`accepted` , steps, heartbeats, then `done` or `error` with a request ID

Stop really stops

the provider call is cancelled and the slot freed

Render-ready

`/ready` readiness and a 20-second drain on SIGTERM



A proxy 502 becomes a Thai message; the text returns to the composer (browser suite 10/10)

What has been measured, and what has not

Test	Result	Scope
10 questions · 5 images · 5 safety cases	7/10 · 1/5 · 5/5	3.0.1 with Typhoon on Render, reviewed answer by answer
Coursework benchmark A / B / C (OFFLINE)	15/20 each	the real app with model stand-ins; images fail on Tesseract read errors
pytest · browser UAT · upgrade · TH/EN	371 · 36/36 · 10/10 · 43/43	4.0.0-rc3, synthetic data, no outbound calls
Resilience R01–R12 · browser recovery	12/12 · 10/10	deterministic fault suite; not uptime, not Render latency
LIVE_FREE with Typhoon and iApp	not run	preflight blocked: policy not reviewed, free status unverified, no trial keys

Three fixes found in live runs, measured before and after

	Before	Fix	After (rounds 2 and 3)
1	Q01 and Q02 withheld (HTTP 502): citation limit too low, report values attached to package answers	Business sources always citable; stray values dropped; links stripped instead of rejecting	Q01 and Q02 answered and passed in both rounds
2	Q07 quoted the corporate price twice (1,980 instead of 990 baht)	Every amount checked against the catalog; one rewrite, then withheld	990 / 1,490 / 2,390 baht per person, correct
3	Q08 and Q09 answered without searching the knowledge base	Questions naming a test always search; medical claims need medical sources	Sources cited; Q08 still overstated its source

Ready for review; next we measure on the real system

Deploy

- 1 Owner merges into `main`
the Blueprint deploys automatically
- 2 Set secrets in the Render Dashboard
database, data key, provider keys, budget cycle
- 3 Check `/health` and `/ready`
version, commit and readiness

Next

- 1 LIVE_FREE 10 / 5 / 5 with synthetic data
after the owner confirms free status and keys
- 2 Clinical and Thai-reader review
every coursework answer
- 3 Demo clip, three minutes or less



Answers from the clinic's data. Explanations the customer can check.

One FastAPI service on Render, Thai first, every step visible, every model call inside a budget, and no request left hanging.

Code and documentation: github.com/siriponsri/LabClear · reports in docs/report/

นายวัชรินทร์ บัวสอน · นายศิริพล ศรีเฮงไพบูลย์